

A. Personal Statement:

I'm an Assistant Professor of Clinical Neurobiology in the Department of Psychiatry at Columbia University Irving Medical Center (CUIMC) and the New York State Psychiatric Institute (NYSPI). To advance mental health research, I apply my interdisciplinary background in machine learning, multimodal neuroimaging, and neurobiology. My research centers on creating and utilizing diverse brain imaging techniques and AI/Deep Learning approaches to support improved psychiatric disorder diagnostics and treatment. My research encompasses four major areas: 1) Advancing Diffusion Tensor Imaging (DTI) preprocessing and automated quality control methods; 2) Identifying brain biomarkers using neuroimaging for mental disorders; 3) Developing AI/deep learning approaches for psychiatric diagnosis and personalized interventions; and 4) Investigating real-time fMRI and EEG neurofeedback as psychiatric treatment methods. All of my research is focused on improving psychiatric outcomes.

[1] Advancing DTI preprocessing and automated quality control methods: Much of my work has focused on improving DTI data reliability, particularly in clinical populations prone to subject movement. Recognizing that motion artifacts significantly compromise data quality, I developed an automated quality control (QC) pipeline enhancing existing neuroimaging toolboxes such as FSL and Freesurfer's TRACULA. This pipeline incorporates objective quality metrics and novel image-processing techniques, including color-encoded fractional anisotropy (FA) images to assess data integrity. These methods were published in the *Journal of Magnetic Resonance Imaging* and *Journal of Neuroscience Methods*, and presented at multiple *Human Brain Mapping* conferences. Recently, I extended this work to address challenges from diffusion gradient cycling (DGC), a protocol on the GE Premier MRI scanner. While DGC mitigates hardware heating, it increases motion-related artifact susceptibility. I'm developing a deep learning-based method using convolutional neural networks (CNNs) to detect and correct corrupted DWI volumes acquired with DGC, enabling real-time, automated DTI quality assessment for pediatric and clinical research settings.

[2] Identifying brain biomarkers using neuroimaging for mental disorders: These DTI data reliability efforts support identifying robust neurobiological markers of psychiatric disorders. In our study *"White Matter Integrity and Functional Connectivity in Adolescents with a Parental History of Substance Use Disorder (SUD)"*, we employed DTI and resting-state fMRI to examine structural and functional brain connectivity in at-risk adolescents. I led investigations to identify neural indicators of treatment outcomes. *"Association of White Matter Integrity With Executive Function and Antidepressant Treatment Outcome in Patients With Late-Life Depression"* found fractional anisotropy variations associated with antidepressant response, suggesting predictive marker potential. *"Altered Frontal Interhemispheric and Fronto-Limbic Structural Connectivity in Unmedicated Adults with Obsessive-Compulsive Disorder"* identified distinctive structural brain abnormalities in Obsessive-Compulsive Disorder (OCD). *"Structural Neural Markers of Response to Cognitive Behavioral Therapy in Pediatric Obsessive-Compulsive Disorder"* demonstrated specific structural connectivity patterns predicting clinical response to behavioral interventions in youth with OCD. These studies highlight the utility of DTI and resting-state fMRI for uncovering neural underpinnings and identifying treatment efficacy biomarkers.

[3] Developing AI and deep learning approaches for psychiatric diagnosis and personalized interventions: We designed sophisticated AI and deep learning techniques analyzing extensive multimodal datasets, including structural MRI, DTI, fMRI, and clinical measures from psychiatric disorder individuals and healthy controls. A key innovation involves a multimodal imaging Variational Autoencoder (VAE) framework with explainable AI methods, substantially improving psychiatric condition classification performance. For example, in *"Denoising VAE as an Explainable Feature Reduction and Diagnostic Pipeline for Autism Based on Resting State fMRI"*, we utilized a large-scale multimodal dataset (~1100 participants), building predictive models for autism spectrum disorder (ASD) diagnosis through deep learning, demonstrating the approach's robustness in identifying consistent ASD neural biomarkers across diverse populations and scanning sites. Our publication *"Explaining deep learning-based representations of resting state functional connectivity data: focusing on interpreting nonlinear patterns in autism spectrum disorder"* examined latent contribution scores at network levels in ASD and healthy control groups. Both groups shared key network contributors—such as the ventral attention network (VAN) driving latent component 0—but showed distinct differences, including divergent VAN (latent 0) and sensory/somatomotor network (latent 2) scores. This research created an interpretable AI model illuminating common and unique network-level functional connectivity patterns in ASD versus controls, offering deeper autism neurobiology insights. The multimodal VAE frameworks and explainable AI tools we developed advance neuroimaging research by enhancing diagnostic accuracy and model transparency, paving the way for interpretable AI-driven clinical tools and highlighting AI and deep learning's transformative potential in personalized mental health diagnostics and treatment.

[4] Investigating real-time fMRI and EEG neurofeedback as psychiatric treatment methods: My work explores real-time functional neuroimaging approaches—particularly real-time fMRI neurofeedback (rtfMRI-NF)—for modulating cognitive and emotional processes. I conducted a pilot connectivity-based rtfMRI-NF study on emotion regulation in healthy individuals and those with a history of MDD, one of the first to target amygdala-prefrontal connectivity in this population. Preliminary findings were presented at the 2017 Real-time Functional Imaging and Neurofeedback (rtFIN) conference, with a preprint available on arXiv titled *Connectivity-based Real-Time fMRI Neurofeedback Training in Youth with a History of Major Depressive Disorder*. We recently conducted a study decoding neural patterns associated with auditory verbal hallucinations (AVH), providing preliminary data for decoded fMRI neurofeedback (DecNef). This work was presented at ACNP 2021 titled *"Decoding auditory verbal hallucinations based on patterns of neural activity in the temporal cortex in schizophrenia"* with a manuscript under revision. Moreover, I successfully developed an rtfMRI-NF platform for the GE Premier scanner at NYSPI. Since DecNef had never been tested on GE, validation tests showed healthy subjects could unconsciously modulate brain activity to non-AVH patterns in two sessions, with significant within-subject effects from session one. Based on this, we submitted an R61/33 grant as MPI using individualized DecNef protocols to modulate temporal-lobe activation for treatment-resistant auditory hallucinations in schizophrenia. We also developed a decoded EEG neurofeedback platform using Muse2 and Muse S—an open-source, deep-learning platform funded by the CUIT Emerging Technologies Grant as co-PI, with findings presented at rtFIN 2022 and 2024, respectively.

In summary, I maintain a strong record of research productivity, with 52 peer-reviewed publications in leading journals, including *Molecular Psychiatry*, *Information Sciences*, *Biological Psychiatry*, *NeuroImage*, *Neuropsychopharmacology*, *Human Brain Mapping*, and *The American Journal of Geriatric Psychiatry*. This body of work continues to grow, with several additional manuscripts currently under review.

B. Education and Training: I earned my Ph.D. in Pattern Recognition and Intelligent Systems from Shanghai Jiao Tong University, focusing on biometric data analysis methodologies. After being named Outstanding Graduate of Shanghai, I began as an Assistant Professor in the Computer Science Department at East China Normal University, focusing on computer vision research. Motivated by brain understanding interests, I transitioned to neuroscience through postdoctoral work under Dr. Joe Z. Tsien at the Brain & Behavior Discovery Institute at Georgia Health Sciences University (now Augusta University), gaining foundational expertise in neuroscience, computational modeling, and electrophysiology. This sparked interests in applying neuroimaging and AI to psychiatric disorders, leading me to join the Brain Imaging Lab at CUIMC, where I received extensive training in MRI physics, structural and diffusion MRI, resting-state and task-based fMRI, and neuroimaging tool development. I collaborated on several NIH R01-funded projects related to Autism, Depression, OCD, Bulimia Nervosa, and schizophrenia. Through a NARSAD Young Investigator Grant and institutional funding support, I received interdisciplinary training that combined clinical insights with technical development in psychiatric research, equipping me with unique skills, bridging advanced computational methods and translational clinical neuroimaging application expertise.

C. Professional Accomplishments, Leadership, and Impact:

Research Accomplishments: I have served as Principal Investigator (PI), Co-Investigator (co-I), or collaborator on 24 funded research projects, including nine as PI, Co-PI, or Subcontract PI. My research involves multidisciplinary team collaborations conducting translational studies across Depression, Substance Use Disorder (SUD), ASD, OCD, Bulimia Nervosa, and Schizophrenia using multimodal neuroimaging techniques. I led a NARSAD Young Investigator Award project and two CUIMC pilot awards focused on identifying multimodal imaging biomarkers for Depression, served as Subcontract PI and Co-I on multiple NIMH-funded R01 grants, an MPI on one pending NIH R61/R33 grant (scored 31), PI on one pending SFARI, and six pending internal CUIMC grants (three MPI/PI and three co-I). I am actively preparing four NIH grant applications as an MPI or PI and one BBRF Distinguished Investigator Grant as co-I, which focus on advancing neuroimaging and neurofeedback research in psychiatry and developmental neuroscience. Specifically, R61/R33 – *Connectivity-Based Real-Time fMRI Neurofeedback for OCD* (2026–2031), R21 – *Decoded fMRI Neurofeedback for Emotion Regulation in Depression* (2026–2028), R21 – *Connectivity-Based Real-Time fMRI Neurofeedback to Enhance Hippocampal-Cortical Connectivity* (2026–2028), R21 – *Developing a Generalized Deep Learning Framework for Diffusion Tensor Imaging Preprocessing* (2026–2028), and BBRF Distinguished Investigator Grant -*Real-Time Glutamate Neurofeedback Regulation as a Therapeutic Target* (2026-2027).

Leadership Roles: I have held multiple leadership and service roles, including Lead Organizer for the Interdisciplinary Collaboration Club Seminar Series and Co-Leader of the Neuroimaging Journal Club in the Department of Biostatistics in Psychiatry at Columbia University, Workshop Co-Chair for the 2024 International

Conference on Brain Informatics and Technical Program Chair for the 2024 CISP-BMEI Congress. I contributed as a Program Committee member for Brain Informatics (2023), NY Scientific Data Summit (2019), and Real-time Functional Imaging and Neurofeedback (2017). At Columbia's Data Science Institute (DSI), I'm active in the Health Analytics, Computing Systems for Data-Driven Science, and Foundations of Data Science committees. I've been an IEEE Senior Member since 2013 and an Organization for Human Brain Mapping (OHBM) member since 2012.

Editorial and Review Activities: I serve as Associate Editor for *Meta-Radiology* and *Frontiers in Psychiatry—Aging Psychiatry*, was Guest Editor for the 2022 special issue on *Post-Stroke Neural Plasticity*, and review for numerous high-impact journals, including *European Journal of Pediatrics*, *NeuroImage*, *Cerebral Cortex*, *Neuroplasticity*, *Magnetic Resonance Imaging*, and various IEEE Transactions.

Invited Talks: I presented at prestigious institutions on real-time fMRI neurofeedback, machine learning applications in mental health, and advanced neuroimaging techniques, including DecNef presentations at Columbia University's Department of Psychiatry and NYSPI, machine learning innovations at Massachusetts General Hospital and Columbia's DSI, and DTI preprocessing talks at GE HealthCare, Nathan Kline Institute, and Child Mind Institute. I presented on connectivity-based and decoded fMRI neurofeedback at the International Symposium on Decoded Neurofeedback in July 2025.

D. Teaching and Mentorship: I engage deeply in teaching and mentorship, guiding postdoctoral fellows, Ph.D. students, and research assistants while designing courses blending theoretical instruction with practical applications through hands-on projects, interactive labs, and real-world case studies. At Columbia, I have taught graduate courses: *Applied Machine Learning for Image Analysis* (2018–2022) and *Applied Machine Learning for Computer Vision* (2023–2025) for MS students from the DSI and Statistics Department. I have co-taught a course, *Fundamentals of Human Brain Imaging: From Theory to Practice* (Fall 2021) in the Psychology Department and taught *Introduction to Brain-Computer Interaction* (Spring 2018). I actively mentor diverse trainees: **Ph.D. Trainees** (5 candidates on neuroimaging, EEG analysis, and machine learning projects, producing first-author publications and advancing to faculty positions); **Postdoctoral Fellows and Visiting Scholars** (5 postdocs and 1 visiting faculty on neuroimaging preprocessing to machine learning projects); **Research Assistants** (8 assistants on PTSD, OCD, MRI acquisition, preprocessing, and QC projects); **Graduate Students** (39 MS students from Statistics, DSI, Computer Science, Psychology, and Biostatistics on DecNef, DTI data analysis, and rtfMRI-NF systems); and **Undergraduate Students** (on neuroimaging and computer science projects). My trainees author papers, posters, and abstracts, transitioning to academia, industry, or data science careers through hands-on training in data acquisition, processing, analysis, and interdisciplinary collaboration.

E. Ongoing/planned projects: At the NYSPI MRI Unit, I'm integrating a DTI QC pipeline into MRI data acquisition workflows for real-time QC and enhancing rtfMRI-NF pipelines with user-friendly GUIs. Current research applies DecNef targeting emotion regulation in MDD patients and AVH in schizophrenia patients, evaluating neurofeedback therapeutic potential. I also plan to conduct two pilot studies—one focused on connectivity-based rtfMRI-NF to enhance hippocampal-cortical connectivity, and another on EEG data analysis for characterizing neural signatures of vibration augmentation and associated mood improvements. These studies will provide preliminary data for near-future NIH grant applications.

Short-term Goals are addressed in one pending R61/33, six pending CUIMC pilot awards, one pending SFARI grant, and two pilot studies. In particular, the NIMH MPI R61/33 (scored 31) evaluates DecNef's potential to alleviate psychotic symptoms in schizophrenia by targeting AVH neural signatures in the temporal lobe. Two MPI/PI CUIMC pilot awards focus on emotion regulation in MDD patients and AVH in schizophrenia patients, underscoring the promise of neurofeedback as a precision tool for understanding and modulating circuit-level dysfunction in psychiatric disorders.

Long-term Goal is to leverage my interdisciplinary background in AI, computational modeling, and multimodal neuroimaging—including real-time fMRI and EEG neurofeedback—driving mental health research innovations. Obtaining an R01 is challenging in this field, but I am confident I will succeed. Meanwhile, I am a major contributor to many Columbia and NYSPI investigators. I aim to continue to contribute to more personalized and effective mental health interventions. A key component involves creating collaborative research ecosystems centered on cutting-edge neuroimaging methodologies for uncovering psychiatric disorders and treatment neural mechanisms. My work seeks to bridge fundamental neuroscience with clinical application by advancing computational tools and neurofeedback technologies, enhancing both understanding and improved mental health treatment.